EART11200: Natural Scientist's Toolkit





This week we will look at the burning of fossil fuels, and its consequences. Balancing equations is important.

Burning hydrocarbons

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

$C_2H_6 + 3.5O_2 \rightarrow 2CO_2 + 3\,H_2O$

$C_3H_8 + 5O_2 \rightarrow 3\,CO_2 + 4\,H_2O$

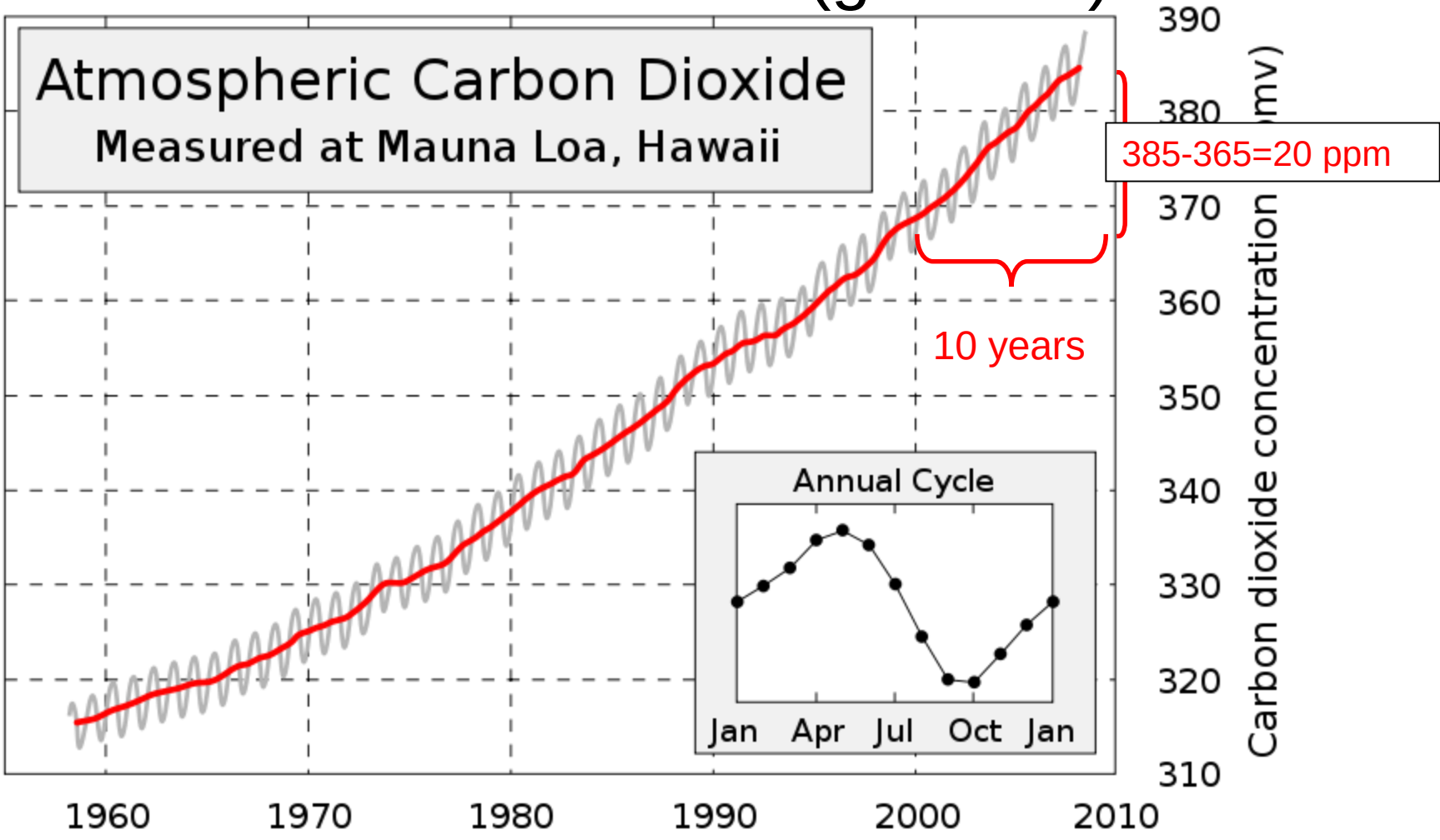
If ***n*** is the length of the carbon chain, the number of hydrogens is ***2n+2***

The number of molecular oxygens must balance with the oxygen on the right, i.e. ***n+(2n+2)/4***

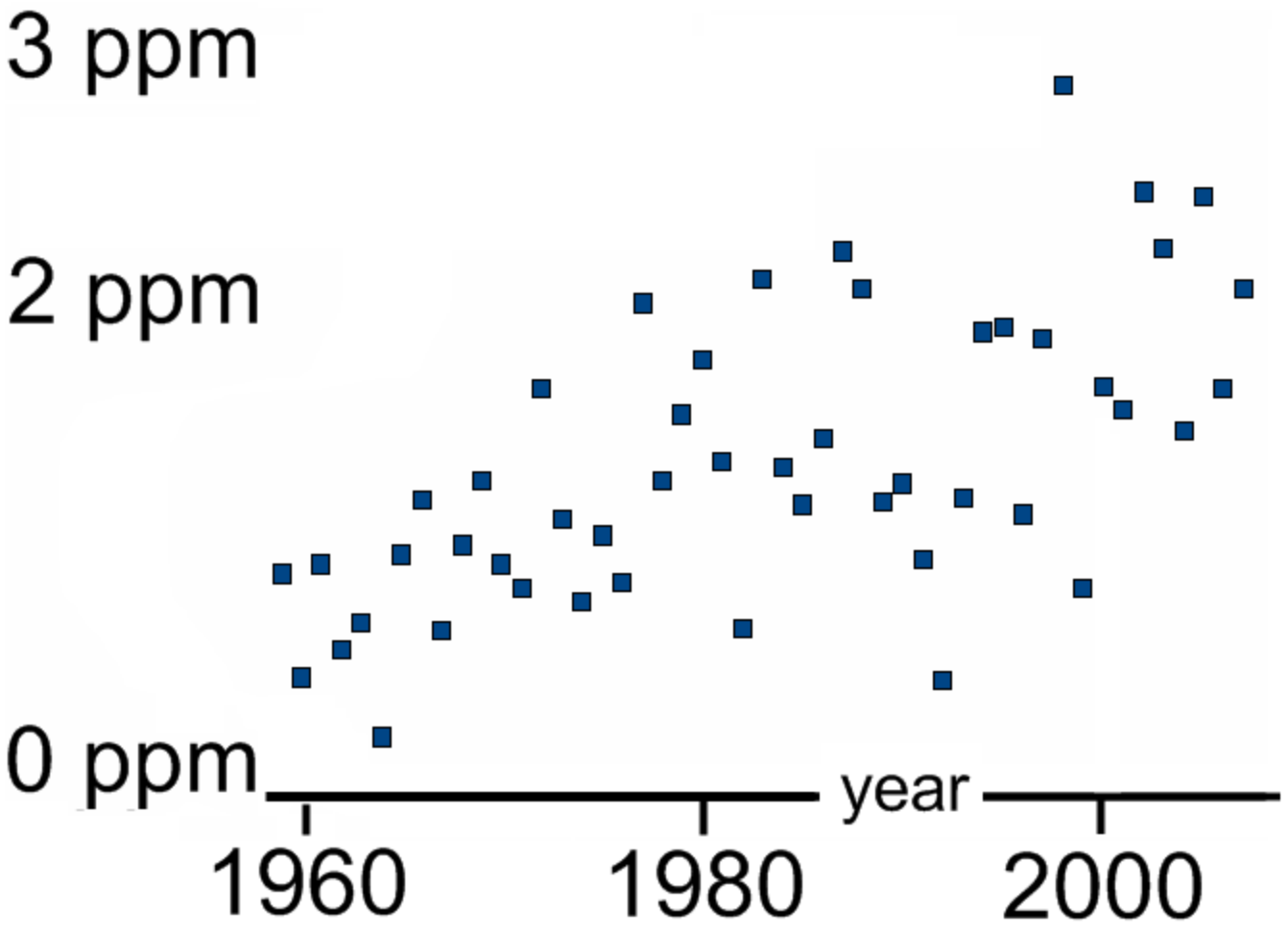
The number of CO2 molecules has to balance the length of the carbon chain i.e. ***n***

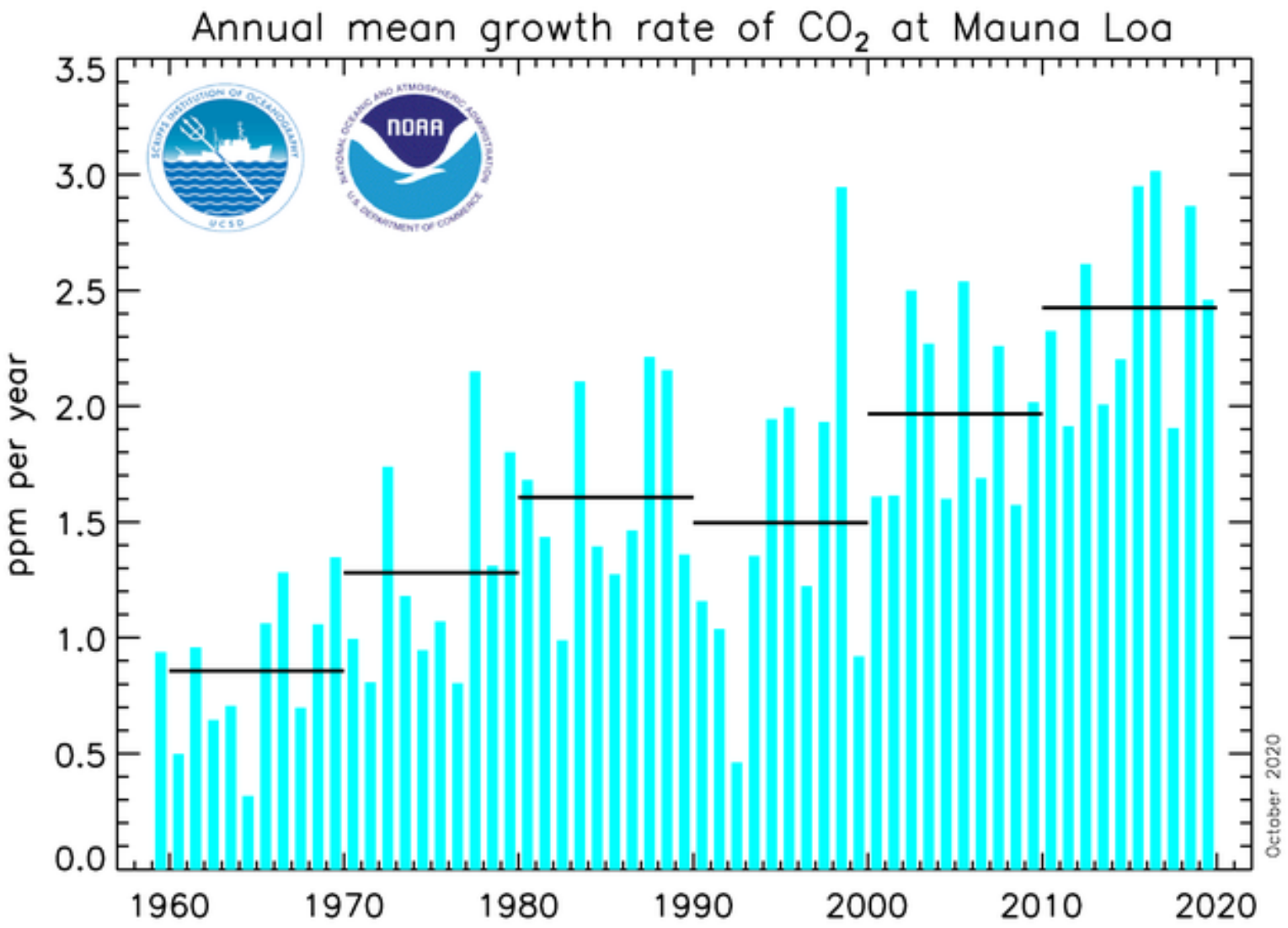
The number of water molecules is determined by balancing the hydrogens i.e. ***(2n+2)/2***

Think about how much the CO_2 concentration has increased over time (gradient)

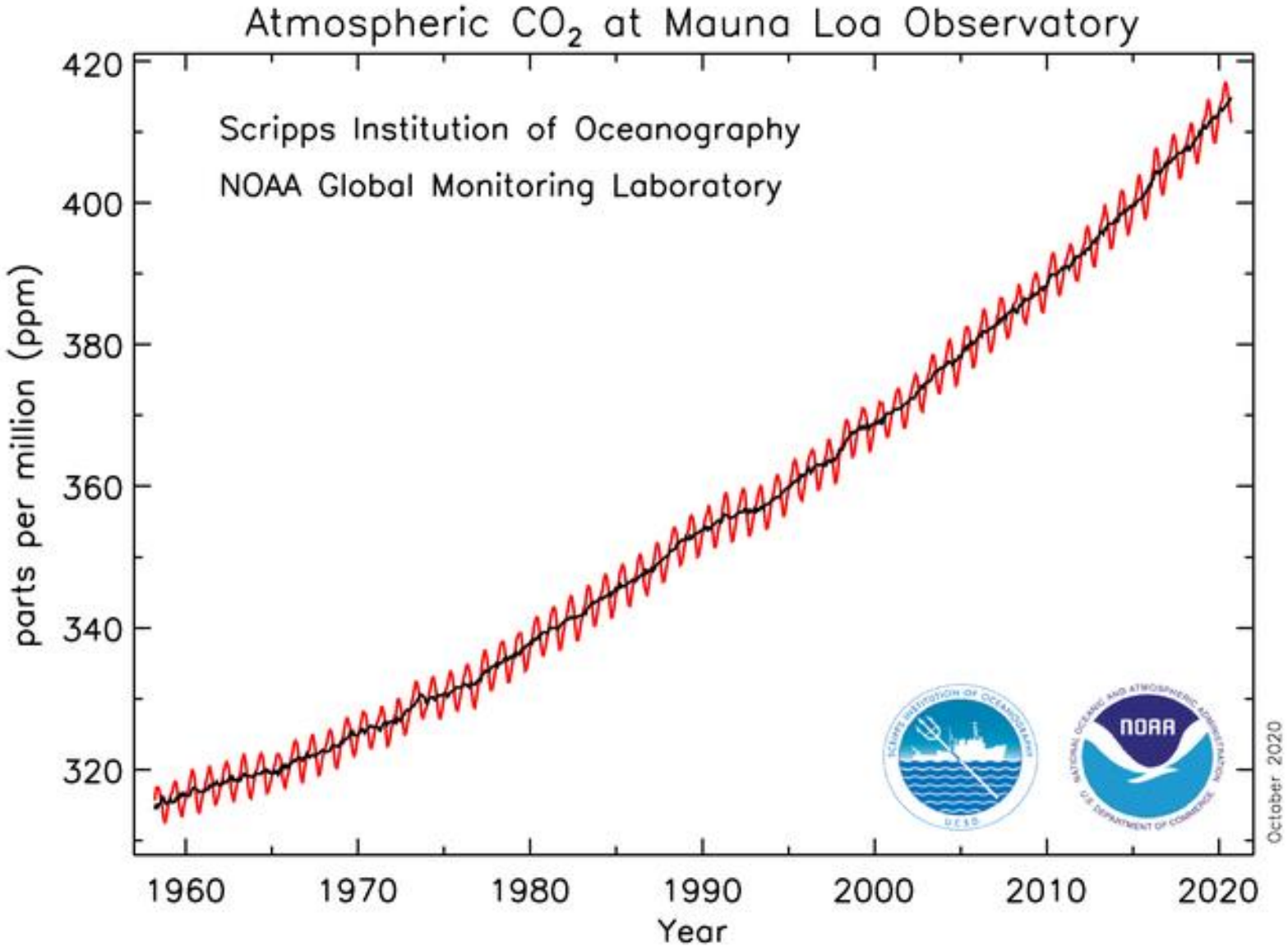


CO_2 change per year

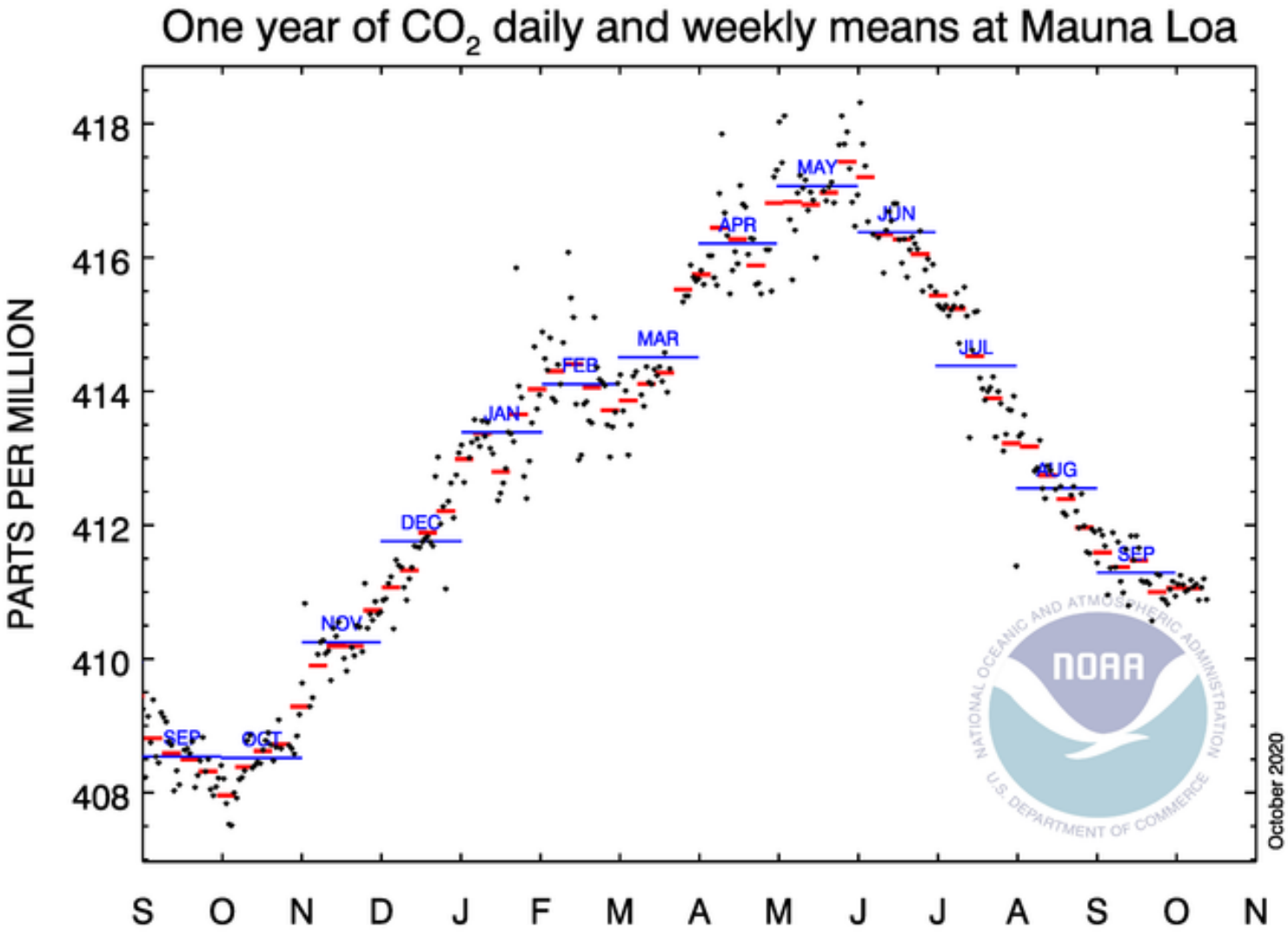




https://www.esrl.noaa.gov/gmd/ccgg/trends/

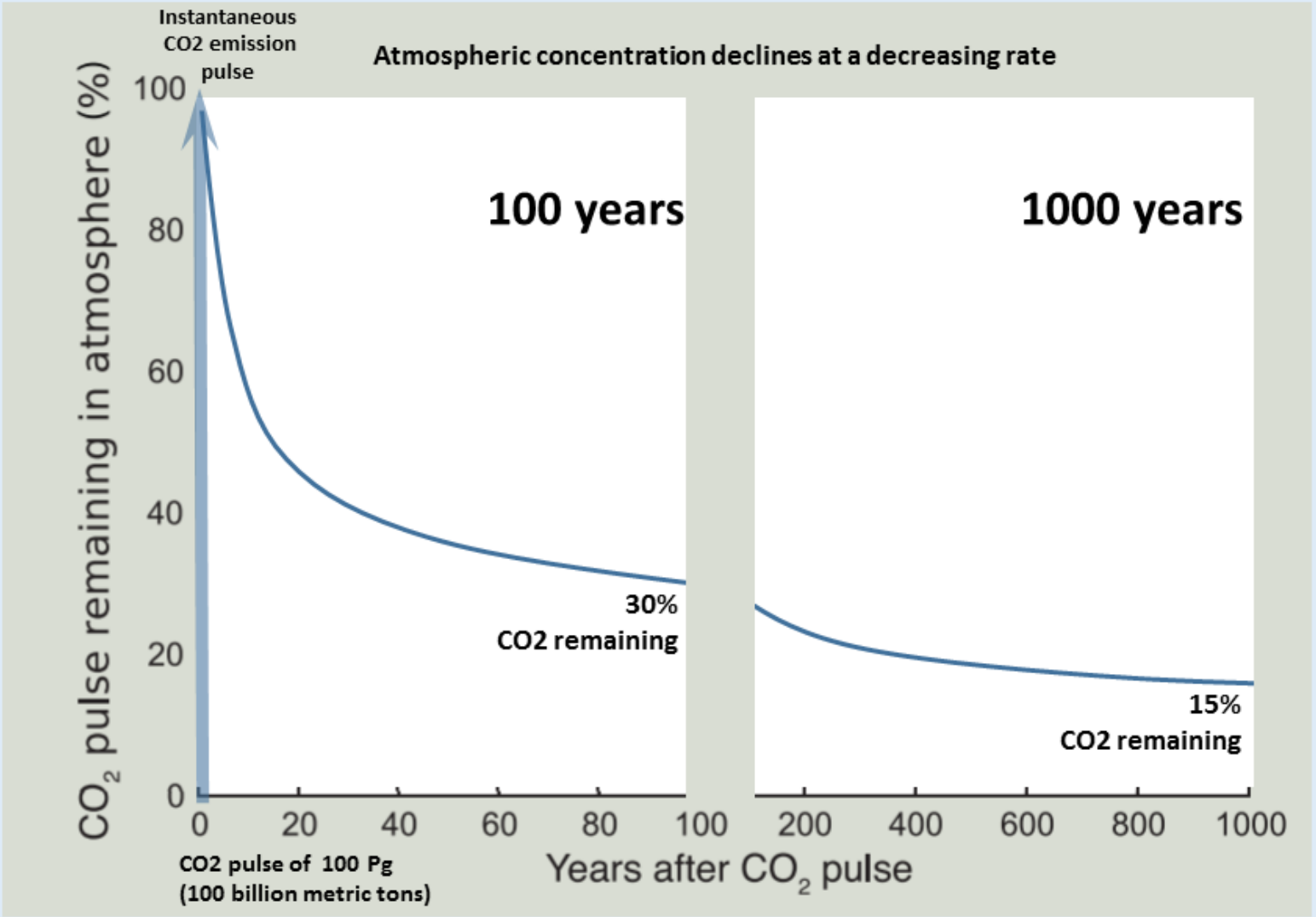


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CO2's ultra long atmospheric residence time



IPCC 2014 AR5 WG1 TFE.7, Figure 1